

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation maintained throughout the project/tabs used throughout	Yes	Improves code readability and consistency.
2	Proper File Structure	Frontend and backend are organized into separate folders with modular architecture.	Yes	Easy to maintain and extend.
3	Meaningful Variable Names	Variables, constants, and objects use descriptive names that reflect their purpose clearly	Yes	Improves code understanding.
4	Function / Method Names	Functions are clearly named according to their functionality.	Yes	Makes the code self-explanatory.
5	Code Comments	Important modules and complex logic include comments for better understanding.	Yes	Simplifies maintenance and debugging.
6	Modular Design	Code is divided into reusable components and API modules	yes	Enhances reusability and scalability.
7	No Redundant Code	Duplicate or unused code has been minimized or removed.	Yes	Optimizes application performance.

8	Error Handling	Exceptions and API errors are handled gracefully with proper messages.	Yes	Improves reliability and user experience.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	Navbar Component	React.js	All frontend pages	High
2	Sidebar Component	React.js	Dashboard and feature modules	High
3	Authentication Service	FastAPI, JWT	Login and Registration	High
4	Axios API Service	JavaScript, Axios	All API communication	High
5	Database Models	SQLAlchemy, SQLite	User, Loan, Financial Profile, History modules	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with clear frontend and backend separation.
Readability	5	Meaningful variable names, consistent formatting, and descriptive function names improve readability.
Reusability	5	Reusable React components, API services, and backend modules reduce code duplication.
Documentation / Comments	4	Key modules and business logic are documented with comments where necessary.
Overall Score	4	The codebase follows modular design principles, is maintainable, scalable, and ready for future enhancements.